RESOLVING A CURVE ON A NOETHERIAN SURFACE BY BLOWUPS IN CLOSED POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file resolve.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the mirrored sequence of blowups of the surface, and the identification of the strict transform with the blowing up of the curve.

Editorial note. The label lemma-blowup-curve names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source resolve.tex –label lemma-blowup-curve –start 4306 –end 4324 –local-id 28123 returned RESULT=OK (L28123 free, occupancy FREE, public interval clear). Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The intrinsic resolution sequence for the curve. Same chapter, the sequence of blowups of the curve itself making it regular. Provides the combinatorial model whose blowups are to be mirrored on the surface.
- Strict transform as the blowing up of the curve. Divisors, the lemma identifying the strict transform of a curve under a blowing up of the surface. Shows that the strict transform of the curve under the mirrored sequence of blowups is the blowup of the curve, hence regular.

Source reference. Original lemma 080E: sources/stacks-chow/tags.txt, source lines 10179–10179. Prerequisite pointer invoked at proof line(s) 4323; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BI4: sources/stacks-resolve/pinned-original-source.tex, source lines 4248–4248. Prerequisite pointer invoked at proof line(s) 4310; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:resolve.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let X be a Noetherian scheme. Let $Y \subset X$ be an integral closed subscheme of dimension 1 satisfying the equivalent conditions of Lemma 0BI4. Then there exists a finite sequence*

$$X_n \to X_{n-1} \to \ldots \to X_1 \to X$$

of blowups in closed points such that the strict transform of Y in X_n is a regular curve.

Proof. Let $Y_n \to Y_{n-1} \to \ldots \to Y_1 \to Y$ be the sequence of blowups given to us by Lemma 0BI4. Let $X_n \to X_{n-1} \to \ldots \to X_1 \to X$ be the corresponding sequence of blowups of X. This works because the strict transform is the blowup by Divisors, Lemma 080E. □